

Single-Panel Model Egitim Raporu

In-Silico Skorlar ILE (v3)

Tarih: 2026-03-24 16:02

Modeller: LightGBM, XGBoost, NN, DNN, SVM

Egitim: General / Hereditary_Cancer | Test-only: PAH / CFTR

1. Veri Ozeti

Toplam veri: 4287 satir

Feature sayisi (FE sonrasi): 989 (target ve Panel haric)

Panel Dagilimi:

General: 3156 satir (pos=2182, neg=974)

Hereditary_Cancer: 715 satir (pos=354, neg=361)

PAH: 324 satir (pos=323, neg=1)

CFTR: 92 satir (pos=92, neg=0)

2. Hold-Out Sonuclari

Panel	Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Sec
General	lightgbm	0.9784	0.9960	0.9982	0.9729	0.9840	0.9292	31
General	xgboost	0.9782	0.9956	0.9981	0.9816	0.9748	0.9299	41
General	nn	0.9612	0.9728	0.9832	0.9590	0.9634	0.8736	210
General	dnn	0.9534	0.9786	0.9901	0.9480	0.9588	0.8470	43
General	svm	0.9668	0.9829	0.9910	0.9679	0.9657	0.8926	142
Hereditary_Cancer	lightgbm	0.9726	0.9924	0.9916	0.9467	1.0000	0.9456	10
Hereditary_Cancer	xgboost	0.9660	0.9919	0.9912	0.9342	1.0000	0.9324	30
Hereditary_Cancer	nn	0.9252	0.9757	0.9750	0.8947	0.9577	0.8483	24
Hereditary_Cancer	dnn	0.9272	0.9714	0.9681	0.8750	0.9859	0.8531	7
Hereditary_Cancer	svm	0.9296	0.9773	0.9759	0.9296	0.9296	0.8601	7

En İyi Model (Panel Bazında):

General: lightgbm (F1=0.9784)

Hereditary_Cancer: lightgbm (F1=0.9726)

3. Test-Only Panel Sonuclari (General modeli ile)

Panel	Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Bal.Acc	N
PAH	lightgbm	0.9969	0.9969	1.0000	0.9969	0.9969	-0.0031	0.4985	324
PAH	xgboost	0.9984	0.9969	1.0000	1.0000	0.9969	0.7060	0.9985	324
CFTR	nn	0.9945	0.0000	0.0000	1.0000	0.9891	0.0000	0.9891	92
PAH	nn	0.9778	0.9814	0.9999	1.0000	0.9567	0.2525	0.9783	324
CFTR	dnn	0.9945	0.0000	0.0000	1.0000	0.9891	0.0000	0.9891	92
PAH	dnn	0.9746	0.9938	1.0000	1.0000	0.9505	0.2365	0.9752	324
CFTR	svm	0.9945	0.0000	0.0000	1.0000	0.9891	0.0000	0.9891	92
PAH	svm	0.9843	0.9907	1.0000	1.0000	0.9690	0.2968	0.9845	324

En iyi Model (Test-Only Panel Bazinda):

PAH: xgboost (F1=0.9984, Recall=0.9969)

CFTR: nn (F1=0.9945, Recall=0.9891)

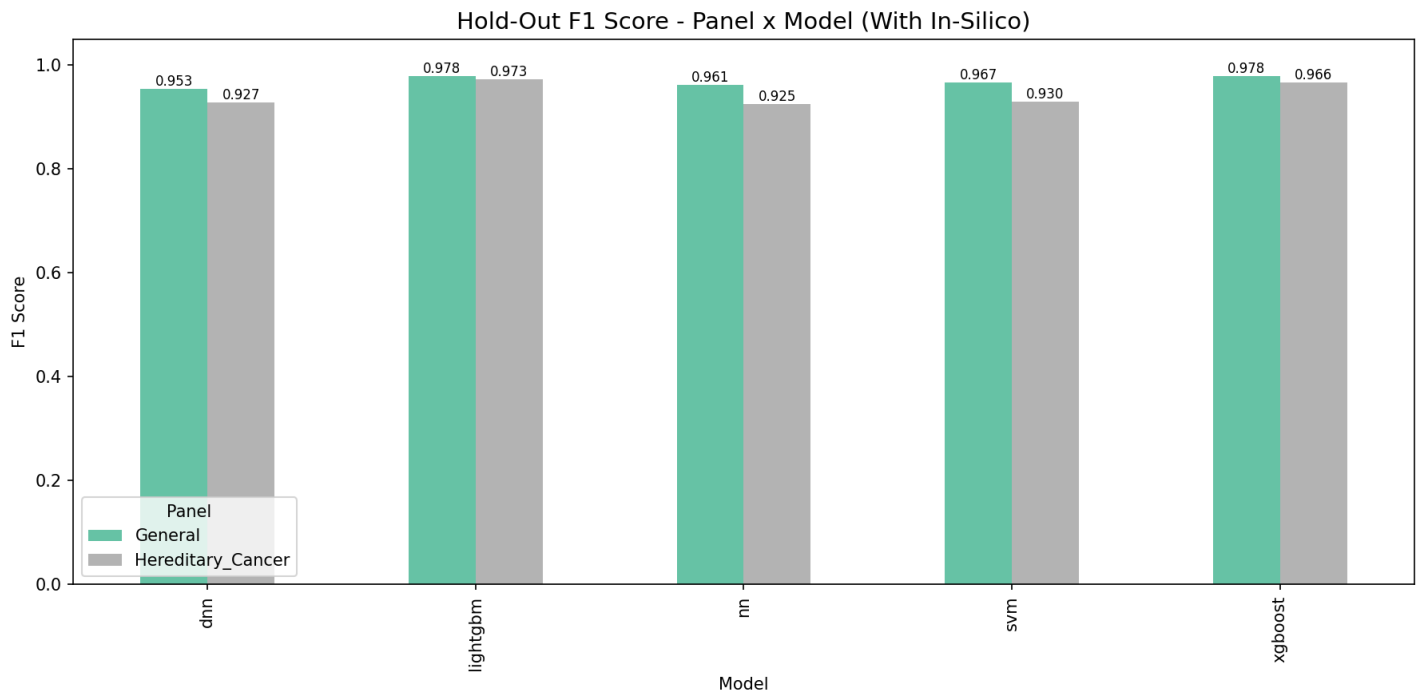
4. Cross-Model Error Analizi

General: 632 test satiri | Tum yanlis: 8 | Cogunluk yanlis: 26 | Tum dogru: 576

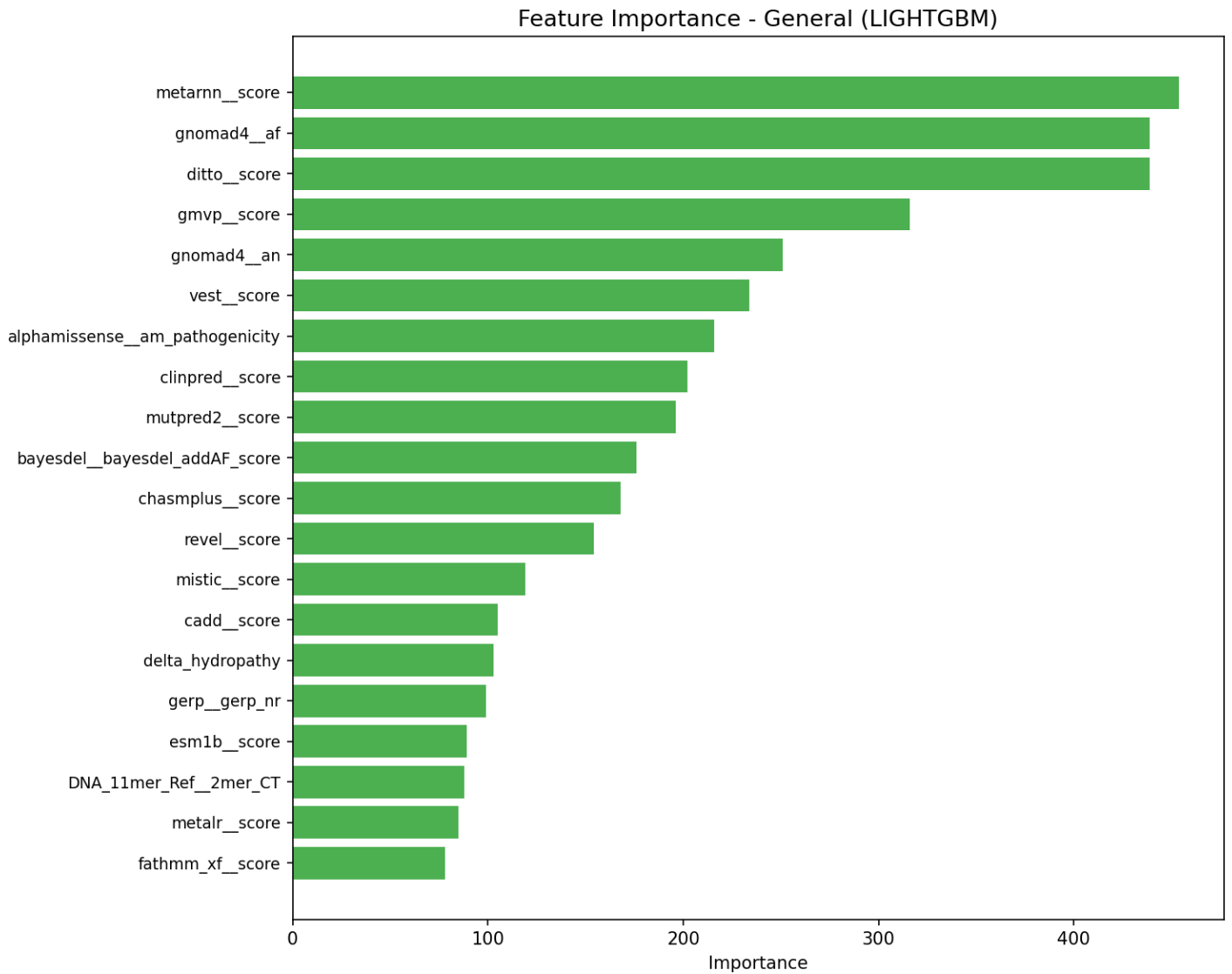
Hereditary_Cancer: 143 test satiri | Tum yanlis: 1 | Cogunluk yanlis: 4 | Tum dogru: 120

Not: Tum modellerin yanlis buldugu satirlar ground truth acisindan supheli olabilir.

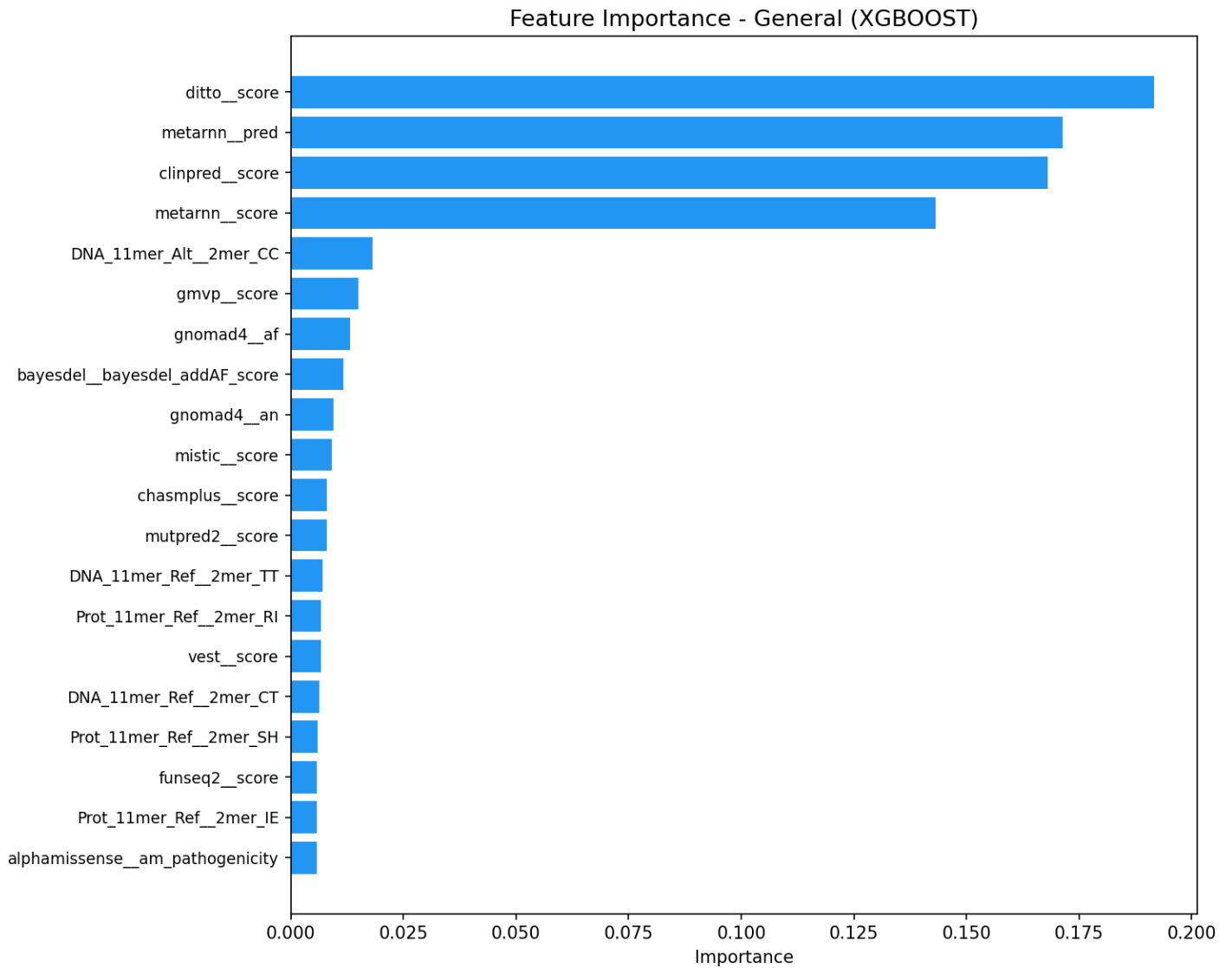
F1 Comparison



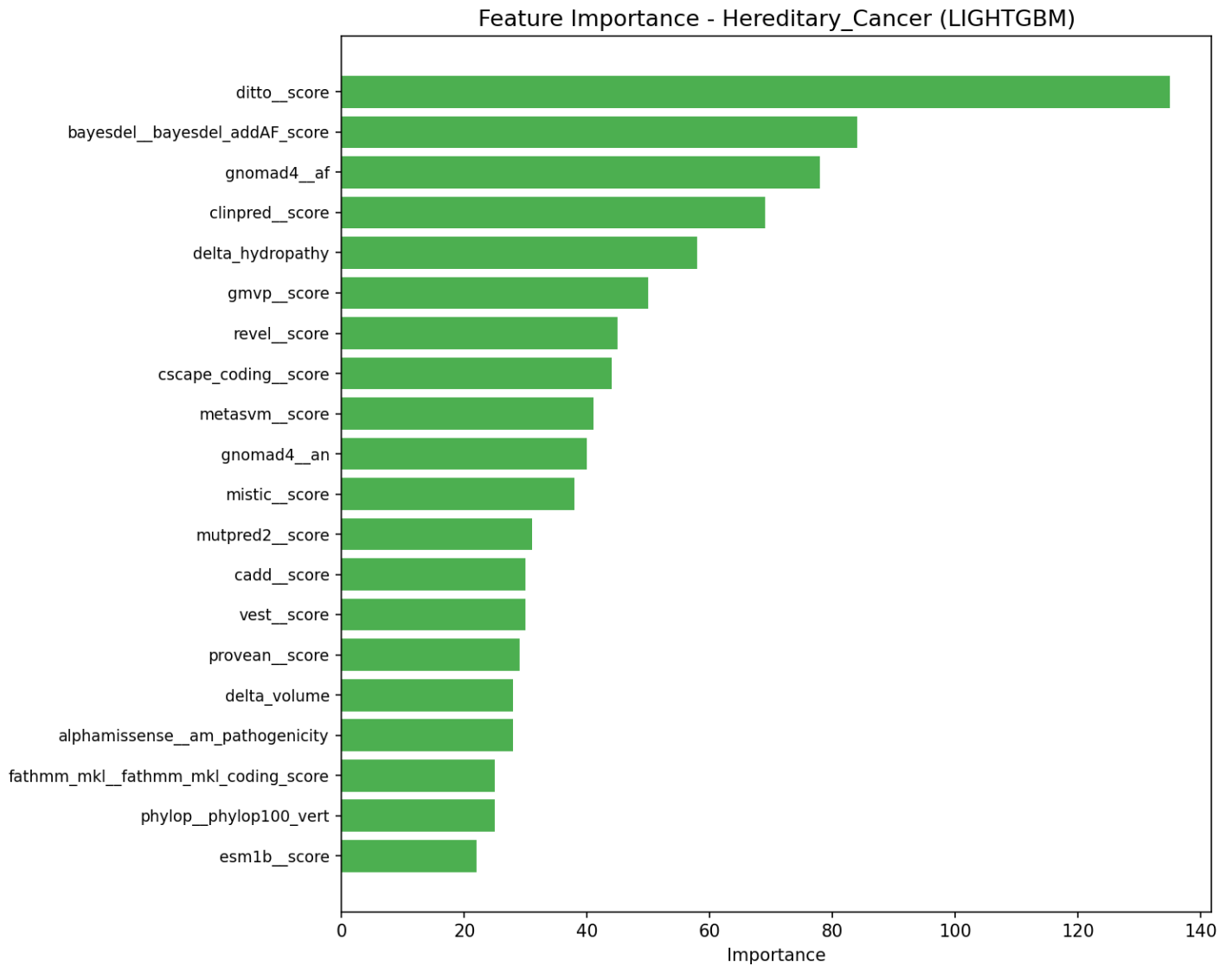
General Lightgbm Feature Importance



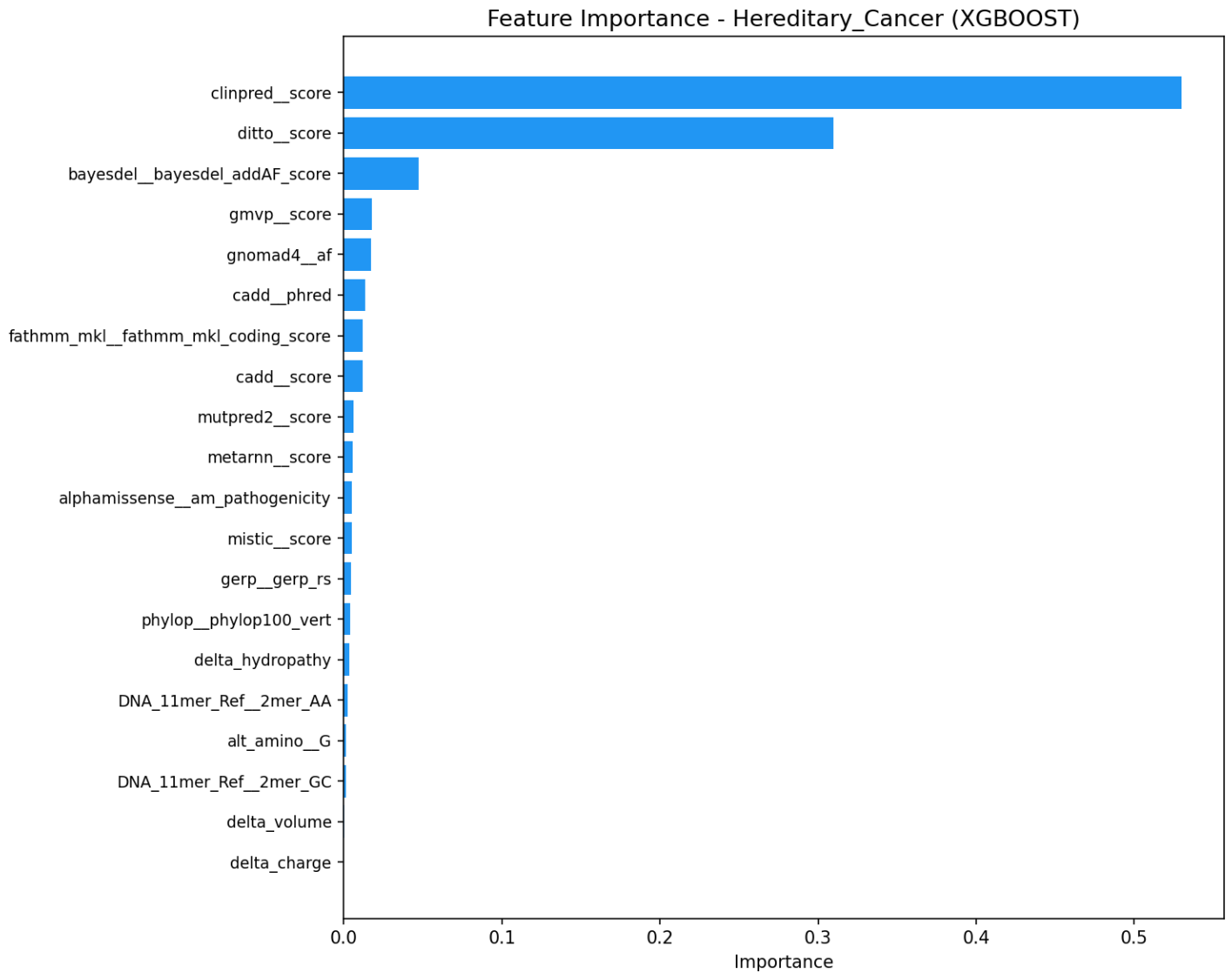
General Xgboost Feature Importance



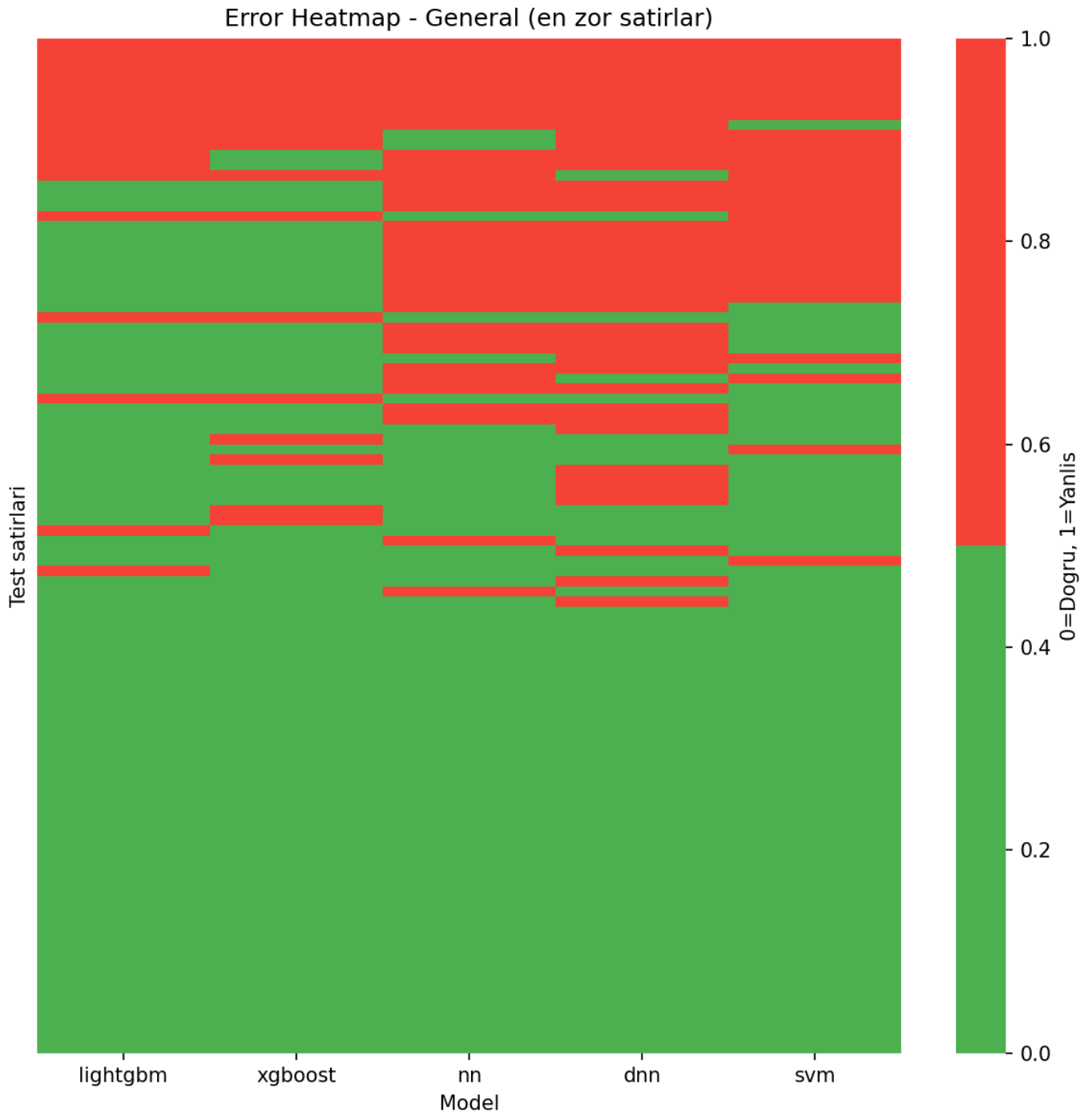
Hereditary Cancer Lightgbm Feature Importance



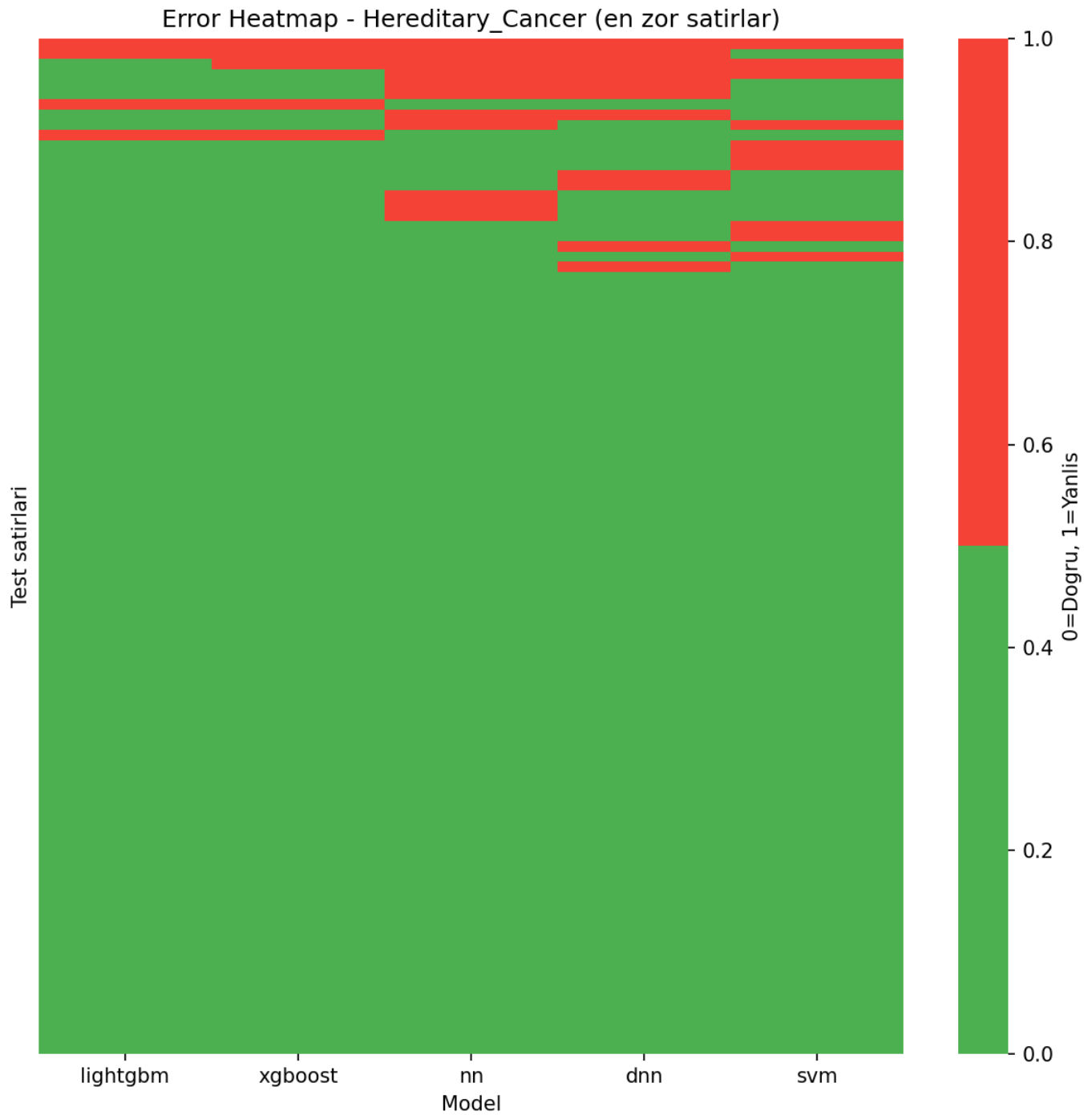
Hereditary Cancer Xgboost Feature Importance



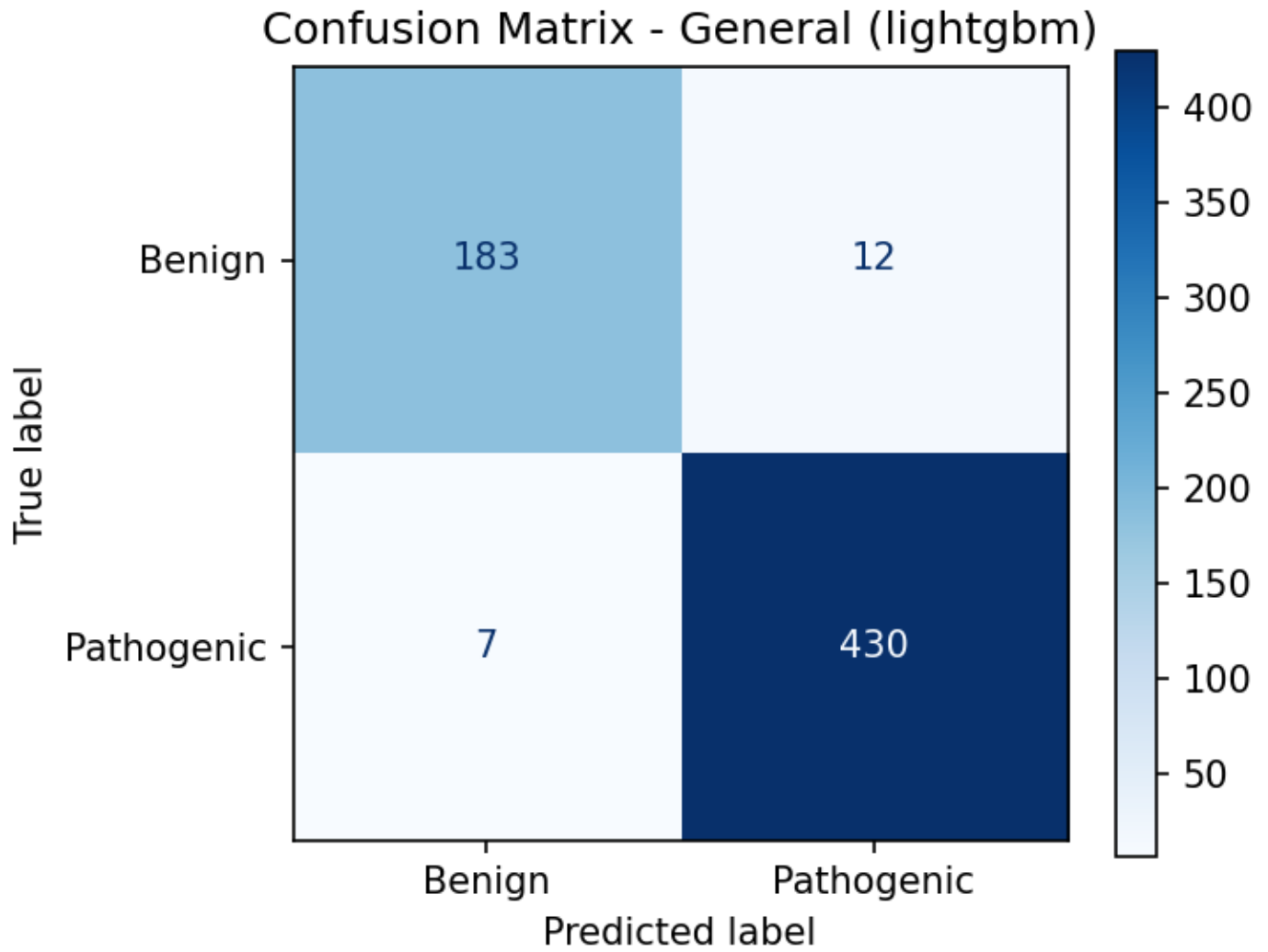
General Error Heatmap



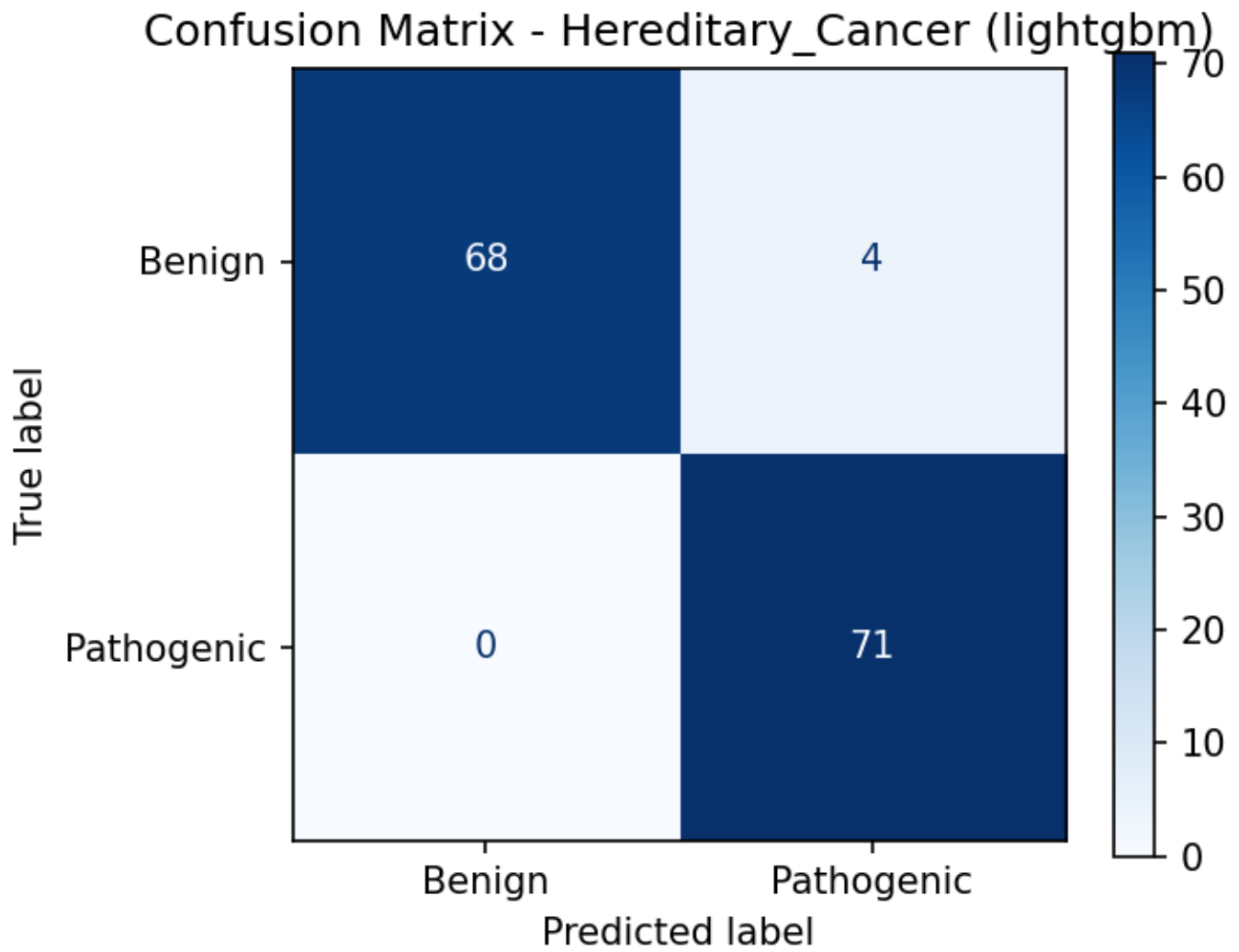
Hereditary Cancer Error Heatmap



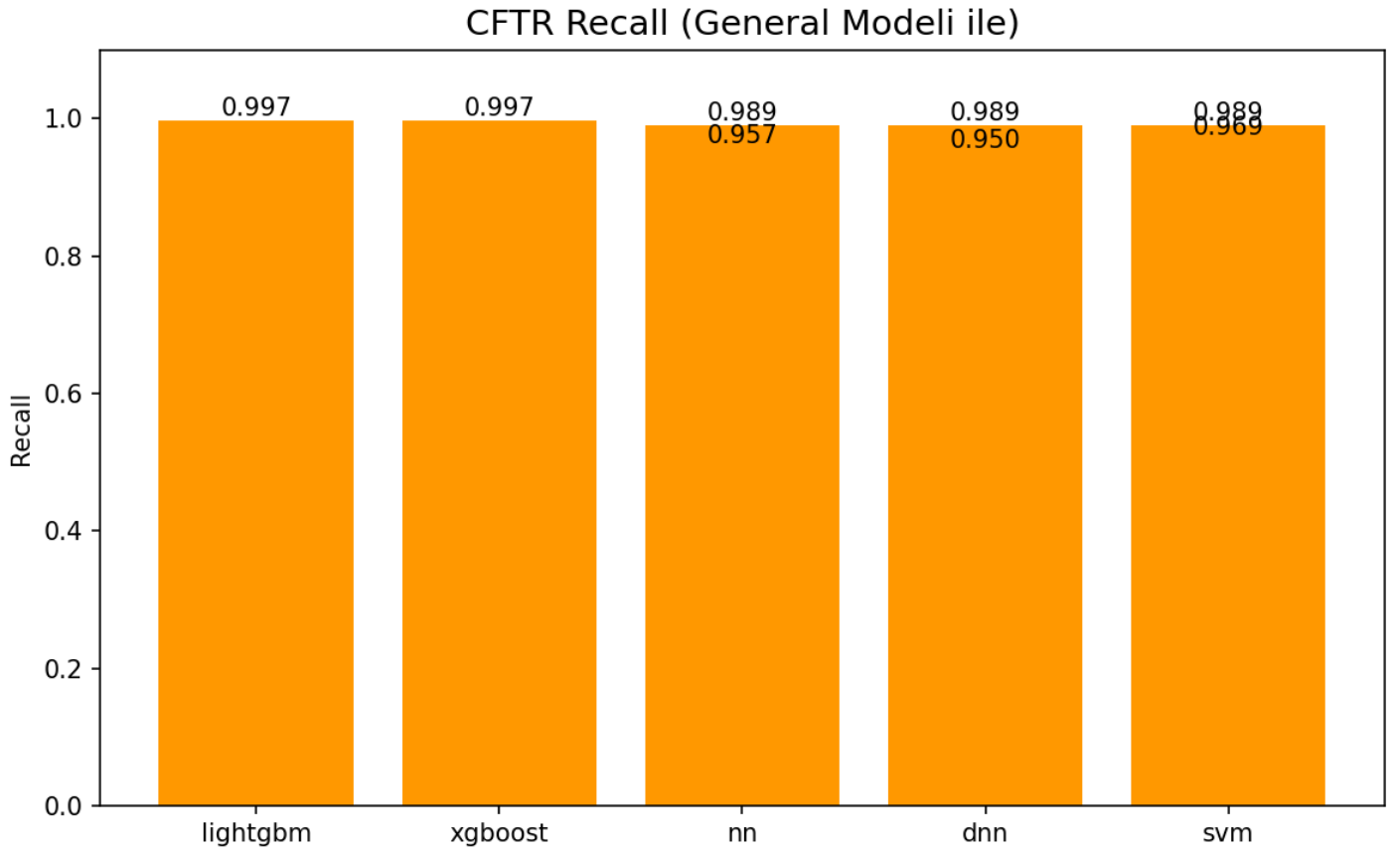
General Confusion Matrix



Hereditary Cancer Confusion Matrix



Cftr Recall



Difficulty Distribution

